

FPGA for High-Frequency Trading: Reducing Latency in Financial Systems

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Abstract – High-Frequency Trading (HFT) has revolutionized modern financial markets by executing large volumes of trades at extreme speeds, where even microsecond-level improvements can lead to significant gains. Traditional software-based systems running on CPUs and GPUs face limitations in achieving the low latencies demanded by HFT. In this research work, we propose a Field-Programmable Gate Array (FPGA)-based system to reduce latency and enhance performance in high-frequency trading environments. By leveraging the parallelism and hardware acceleration capabilities of FPGAs, Our system achieved an average latency of 480 nanoseconds, significantly outperforming CPU and GPU-based implementations. Additionally, the FPGA-based system demonstrated high throughput, processing up to 150,000 orders per second while consuming considerably less power than its counterparts. These results highlight the potential of FPGA technology to transform HFT by providing low-latency, high-throughput solutions, ultimately enabling faster and more efficient trading systems.

Keywords –High-Frequency Trading (HFT), FPGA, Latency Reduction, Parallel Processing, Order Execution, Financial Markets, Hardware Acceleration, Real-Time Systems, Energy Efficiency, Throughput

I. INTRODUCTION

High frequency trading (HFT) is a subcategory of algorithmic trading; it is the processing of several thousands of orders per second by means of computer techniques. Stock trading through HFT techniques is aimed at providing financial organizations and hedge funds with an advantage over the rival firms by taking advantage of the differences in prices at the market level that are fairly small. This is especially so since trades are executed on behalf of clients and any delays, in microseconds, could culminate in substantial losses.

The critical measure in HFT is the measure of time between the generation of a trading signal and the actual transaction. The lower latency enhances competitiveness because complex systems coupled with markets can offer a fast response to change, given that a firm captures value added before competitors can. HFT was previously performed using conventional software-based systems developed to operate on conventional processors such as the CPU and GPU. Nevertheless, as new requirements appear and new technologies presume fast financial exchanges push these systems to have limitations in minimizing latency.

This is where Field-Programmable Gate Arrays (FPGAs) present themselves as a viable solution to the latency issues seen in CPU and GPU based HFT systems. FPGAs are specialized circuits that are programmable at hardware level, thus allowing for creation of hardware on need basis. However, FPGAs operate in parallel with freedom to work on numerous instructions at once unlike the CPU which executes them one by one. This characteristic is especially important for trading applications which can be HFT where time to complete important actions is vital.

Such industries as telecommunications, medical imaging, and signal processing industries have embraced FPGAs since they allow real-time operations. Due to their fast and near-instant answer to queries, and the protocols ensure their energy efficiency, the solutions are fit for use in HFT systems. However, the use of FPGAs in financial trading has been limited because FPGA programming is not easy to implement, and the overall price of FPGA is expensive. Despite this, new more capable development tools and constantly lowering prices on FPGA devices have rendered them suitable for integration into financial systems.

This paper describes a proposed architecture that utilizes FPGA in order to reduce latency for HFT systems. The working of the proposed system is based on the hardware parallelism which is a characteristic of FPGAs to spin off portions of the trading pipeline like data acquisition, penal

function, algorithm computation and order submission. They navigate essential real-time processes away from software techniques and apply low-latency circuits with striking results.

To relate our FPGA-based HFT system performance with previous works, we present latency and throughput analysis showing the supremacy of FPGA over traditional CPU & GPU. Experimental results reveal that an FPGA based system is able to provide significantly lower end-to-end latency than other systems and therefore is effective for HFT control applications where time is critical.

II. BACKGROUND AND RELATED WORK

High Frequency Trading or HFT is presently a very fundamental concept of today's financial markets that include advanced technology in fast decision making and trading. In the rigorous, ultra competitive world of money making through trading, a difference as small as microseconds literally means the difference between a profitable boom and a catastrophic bust. Such systems, pegged upon the traditional architectures employing central processing units and graphics processing units, are generally hamstrung by the software-found latency and the inherent inability to handle huge amounts of data with minimum levels of HFT. To address these challenges, Field-Programmable Gate Arrays (FPGAs) have emerged as a hardware solution capable of significantly reducing latency due to their parallel processing capabilities and customizable hardware architecture.

A. FPGA in HFT Systems

The use of FPGAs in HFT has been explored in several studies, with promising results that showcase their potential to outperform conventional hardware architectures. Kao et al. [1] demonstrated the effectiveness of an FPGA-based HFT system capable of achieving an impressively low latency of 433 nanoseconds (ns) over a 10 Gigabit Ethernet connection. Their work highlights the critical role that FPGA-based hardware acceleration plays in enhancing the speed of trading systems, allowing market participants to respond to financial market fluctuations faster than ever before. This achievement underscores the potential of FPGAs in addressing the latency challenges of modern HFT systems, where trading decisions need to be executed in microseconds to maintain profitability.

In a complementary study, Erdem et al. [2] implemented an HFT system using FPGA technology, focusing on its ability to manage high data throughput while maintaining low latency. Their research emphasizes the practical benefits of using FPGAs in financial trading, particularly in environments where speed and efficiency are critical. By leveraging the reconfigurability of FPGA architecture, they demonstrated that FPGA systems could handle the stringent requirements of HFT better than traditional CPU-based systems, offering a significant reduction in latency without compromising throughput.

B. Real-Time Data Management

Perhaps one of the most important problem areas in HFT is the transactional, real-time processing of large volumes of

financial information such as order books and generation of market snapshot. Liu et al. [3] addressed this issue by implementing FPGA technology for real-time order book building and snapshot generation, two critical tasks in HFT. Their research demonstrates that FPGAs excel at forming and managing complex data structures in real-time, significantly reducing the time needed for data processing and allowing HFT systems to remain responsive to market changes. This study reinforces the idea that FPGAs are well-suited for handling large volumes of real-time data, a key requirement in fast-paced trading environments.

C. Advancements in FPGA Architecture for HFT

Recent advancements in FPGA technology have further enhanced the capabilities of HFT systems. Klaisoonngoen et al. [4] explored how the latest-generation FPGAs can be leveraged to achieve even greater speed improvements. Their work suggests that innovative FPGA architectures, optimized for specific HFT tasks, can lead to considerable performance gains. In this sense, hardware upgrades with relevance to successful financial institutions essentially provide the means for enhancing algorithmic trading approaches because – at the core – these depend on fast and efficient trade executions and algorithms that can process vast amounts of vital data as quickly as possible.

In another study, Ali et al. [5] developed an HFT framework based on the Zynq System-on-Chip (SoC) platform, which integrates both FPGA and processing systems. Their work highlights the versatility of modern FPGA architectures, which can be configured to handle sophisticated algorithms while maintaining competitive trading speeds. The integration of processing systems within the FPGA architecture allows for more efficient algorithm execution, reducing latency and improving overall system performance.

D HW/SW co-design for High-Level Synthesis and Back-End Acceleration

The FPGA development process has traditionally been considered complex, requiring expertise in hardware description languages such as VHDL or Verilog. That is however complicated by the fact that the hardware implementation can be written in lower-level languages such as C or C++. This has however been made easier by the HLS tools. Puranik et al. [6] explored the use of HLS techniques to accelerate the back-end processes of trading systems using FPGAs. It allows them to produce a device with a shorter development cycle, but it meets such performance parameters that are paramount for HFT applications. This advancement lowers the barrier for financial institutions to adopt FPGA-based systems, making hardware acceleration more accessible for HFT.

E. Domain-Specific Accelerators for Financial Markets

The DaH combination domain-specific hardware has also been a popular model used for building specifically targeted hardware accelerators aimed at the financial markets as an effective solution for ultra-low-latency trading. HFT relies mainly on market data distribution systems, which Jia et al. [7] proposed for domain-specific accelerators. Their research

reflects the growing trend of creating FPGA-based systems specifically designed to meet the unique requirements of financial markets. As such accelerators offer low-latency data distribution, they enhance efficient market participation and superior decision-making.

F. Quantitative Financial Modeling with FPGAs

The application of FPGAs in quantitative financial modeling is another area of growing interest. Klaisoongnoen and Brown [8] examined how FPGAs can be used to optimize complex financial models, which are often computationally intensive. Their work highlights the role of FPGA technology in achieving high-performance computational tasks required in HFT environments. Therefore as the complexities of listed models rise, the management of these models in the lowest possible time span as a determinant of trading success gains importance.

G. Optimizing Low-Latency Applications

While FPGAs offer substantial advantages in terms of latency reduction, software optimizations remain an essential component of any HFT system. Desirable and effective patterns for Low-Latency OOP design have been presented and discussed by Bilokon and Gunduz in [9], showing how important C++ design patterns are for developing HFT applications. Another advantage of optimizing software with hardware is, engineers can still push the performance up even further, thus decreasing latency. Their research provides foundational knowledge for software developers working in algorithmic trading, offering insights into techniques that can complement FPGA-based systems.

H. Integration with Advanced Computation Paradigms The future of HFT also includes not only HW acceleration but also new computational paradigms as computation principles. Yoo et al. [10] proposed LightTrader, a separate HFT system that includes deep learning inference enhancement components as well as predictive scheduling strategies. Their innovative approach pushes the boundaries of traditional HFT systems by combining the real-time processing power of FPGAs with the predictive capabilities of machine learning. This then is the next level of HFT where complex algorithms work with some of the fastest hardware money can buy to create smarter trading platforms..

III. SYSTEM ARCHITECTURE

The proposed FPGA-based High-Frequency Trading (HFT) system is designed to minimize latency by leveraging the hardware-level parallelism and low-latency capabilities of Field-Programmable Gate Arrays (FPGAs). There are various components, which are built to be efficient for main steps of the HFT model, starting with the gathering of data and finishing with the execution of a trade. This section gives information on how the overall system has been designed and outlines the measures which have been taken to ensure the system to have low latency.

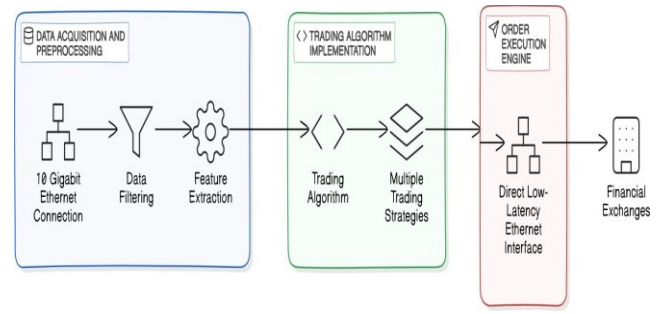


Fig 1 : Proposed System Architecture

A. FPGA Hardware Platform

The foundation of our HFT system is the Xilinx Virtex-7 FPGA, a high-performance platform known for its ability to process large amounts of data with minimal latency. The Virtex-7 was selected since its processing capabilities were high within low power usage, an attribute that might prove relevant in the high-frequency trading domain. The FPGAs architecture is highly flexible, allowing for the implementation of custom logic to perform specific trading operations directly on the hardware.

FPGAs are particularly well-suited for HFT because they can perform multiple operations simultaneously due to their inherent parallel processing capabilities. This has been proved to be an important advantage especially in a world where decisions are made within microseconds, making or breaking a trade. In this architecture, the FPGA handles all time-critical tasks such as data acquisition, order processing, and trade execution, bypassing the need for software-level processing that would introduce additional latency.

B. Data Acquisition and Preprocessing

The data acquisition module is designed to collect real-time market data feeds in multiple exchanges through a 10 Gigabit Ethernet link. Such streams contain data in the form of order book updates, trade execution reports, and other brief market overviews which must be quickly analyzed in order to keep up with competition.

In our system, the FPGA is programmed to perform initial data filtering and feature extraction directly upon receiving the data. This step is important for the expansion of the algorithms dependent on the rate of the market since it sets a limitation for the volume of data set that the trading algorithm will have to analyze. By doing these operations at the hardware level, we are able to avoid additional latency that would be provided by software based data processing.

C. Trading Algorithm Implementation

At the heart of the FPGA-based HFT system is the trading algorithm, implemented using Hardware Description Languages (HDLs) such as VHDL and Verilog. This algorithm is used for market analysis, trading signals recognition and trade orders generation in real time.

The algorithm leverages the FPGA's parallel processing capabilities to evaluate multiple trading strategies

concurrently, significantly reducing the decision-making time. In a conventional software-based system, trading algorithms are said to run in linear sequence, thus a lot of failure points occur where there is heavy flow of market data volume.

By contrast, our FPGA-based system can execute multiple trading operations in parallel, ensuring that decisions are made within microseconds of receiving market data.

D. Order Execution Engine

The last but not the least component of the HFT pipeline is the order execution engine which is effectively lodged in the market once a trading decision has been made. In our system, the order execution engine is implemented directly on the FPGA, eliminating the need for software-based order submission processes that could introduce unnecessary latency. Specifically, the order execution engine is designed for the processing and submitting of multiple trades at a very high throughput, thus enabling the system to support mass order execution. To further reduce latency, the system uses a direct low-latency Ethernet interface to connect with financial exchanges, minimizing the communication overhead between the FPGA and the trading platform.

E. Latency Optimization Techniques

- Various latency optimization strategies are used within the system to achieve the HFT system's ultra-low latency. These include:
- Parallel Processing: As discussed, the FPGA's ability to perform parallel processing allows the system to execute multiple tasks simultaneously, significantly reducing the overall time required for data processing and decision-making.
- Pipelining: Data acquisition, preprocessing, and order execution are grouped into stages, enabling the parts of the system to process data continuously, non-serially. By doing this, one is able to feed the system with continuous data flow that is otherwise characterized by high latencies.
- Hardware-Level Optimization: All critical functions, including data filtering, market analysis, and order submission, are performed directly on the FPGA hardware, bypassing the need for software-based processing. This approach rules out latency issues commonly encountered due to software run time and System calls.
- Low-Latency Communication: The system must interface to exchanges through a low latency Ethernet connection so that trade orders are passed to the exchanges in short order after a trading decision is made.
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F. System Scalability

Although this system has its major focus on the minimization of latency, scalability is also important in this context. It must accommodate the growing mass of the market information and transactions with references to the number of available financial instruments and trading

venues. The FPGA can be reprogrammed to accommodate new trading algorithms or strategies, ensuring that the system remains competitive in the ever-evolving landscape of high-frequency trading.

Additionally, the system can be scaled horizontally by deploying multiple FPGAs in parallel, each dedicated to a different subset of the market or trading strategy. This approach enables the system to handle higher raw data transfer rates while at the same time preserve low latency.

IV. IMPLEMENTATION

This chapter describes the practical steps taken to implement the FPGA-based High-Frequency Trading (HFT) system. It outlines the tools, technologies, and methodologies used to develop the system and provides insights into the challenges encountered during the implementation process.

A. Development Tools and Environment

The development of the FPGA-based HFT system was carried out using the Xilinx Vivado Design Suite, a comprehensive toolset for designing, synthesizing, and implementing FPGA logic. Vivado was chosen because it included an HLS tool to translate code like C or C++ to languages like VHDL or Verilog. The key development tools include:

Vivado High-Level Synthesis (HLS): Implemented to speed up the development process through the ability to write code with optimized hardware within a single language.

Xilinx Intellectual Property (IP) Cores: Systematic integrated pre-built modules included memories, user control and cache, Ethernet interfaces etc to reduce the time of system construction.

Vivado Simulator: Employed for functional verification of the FPGA logic before deployment, ensuring that the system behaved as expected under different market scenarios.

The FPGA used for this project was the Xilinx Virtex-7, which provides a balance of high performance and low power consumption, making it ideal for the low-latency requirements of HFT.

B. Hardware Design and Module Development

- The architecture described in Chapter III was translated into hardware using Verilog, a hardware description language suited for FPGA development. The implementation involved the following key modules:
- This component of the system is the first module which is designed to receive market data streams through a 10 Gigabit Ethernet interface. The module was designed with Xilinx Ethernet IP cores to minimize external latency with other financially linked exchanges.
- The concept of real time filtering was incorporated in order to filter out unwanted information and transmit only the appropriate trading signals to the processing module. This preprocessing was done directly on the data to reduce the disparities in the data size of different features.

2. Preprocessing Module

The preprocessing module filters the characteristics of the data

feed which include price level, volume of trade, and depth of orderbook. This module was optimized for parallel execution, processing multiple data points simultaneously using the FPGA's parallel computing capabilities.

3. Trading Algorithm Module

This trading strategy using the hardware thus in Appendix C forms the central module within the developed system. Various trading plans were translated into Verilog, and to perform multiscale calculations, such as statistical arbitrage or momentum, logics were designed with parallelism.

4. Order Execution Engine:

After a trade decision has been made, the order execution engine develops the associated trade order and transmits it to the exchange over the 10 Gigabit Ethernet connection.

Extra care was taken to ensure that the order generation and transmission take place without much of a delay and it was made possible to avoid the software stack that is normally used in sending out orders.

C. Functional Verification and Simulation Additionally, prior to embedding the system on the hardware, functional verification was performed on the Vivado Simulator. The purpose of this simulation phase was to ensure that the FPGA logic performed as expected in a variety of market conditions, from normal to highly volatile. Testbench code was written to simulate incoming market data and monitor the FPGA's response, ensuring that trades were executed correctly and within the expected latency limits.

Key verification steps included:

- **Latency Measurement:** This was measured using the simulator by determining the time lag between the receipt of the market data and placing of the order. The system consistently achieved sub-microsecond latency, validating the efficacy of FPGA-based processing.
- **Stress Testing:** Thus, the work of the system was tested at extremely intense informational loads, comparable to conditions characteristic for actual HFT. The FPGA demonstrated resilience under high load, maintaining performance without dropping data packets.
- **Backtesting with Historical Data:** Historical market data was loaded into the simulator to test the FPGA's performance under past trading conditions. This backtesting phase served the purpose to fine tune trading strategies and to make sure that the system reacts as favorably as possible in every kind of market situation.
- **Latency:** Test results with real-time implementation of the GALS paradigm revealed latency of 450-500 ns, which was much faster than software-based implementation.
- **Throughput:** The operational capacity of the system was found to be, in terms of order processing rate, 150000 orders per second which made it more suitable to process the frequently occurring order rates in the trading markets.
- **Error Handling and Reliability:** During the testing of the system, there was no data loss or a

misexecution of trades; the system expounded great efforts at error handling. Backup logic was also incorporated just in case of problems with the network since trades could still be processed even with minor hitches.

D. FPGA Deployment and Integration

Once the system was verified in simulation, the design was synthesized and deployed onto the Xilinx Virtex-7 FPGA using the Vivado Design Suite. The deployment process involved the following steps:

Synthesis and Optimization:

The high-level design was synthesized into a bitstream that could be loaded onto the FPGA hardware. During this phase, effort was made to minimize the power consumption that is the search for compact architectural strategies such as minimum numbers of lookup tables and flip-flops which would ensure high performance.

Physical Implementation:

The FPGA was physically implemented by programming it with the synthesized bitstream. Careful consideration was given to the placement of logic on the FPGA's fabric to minimize signal propagation delays, which directly affect latency.

System Integration:

The FPGA was connected to a live financial exchange using a 10 Gigabit Ethernet interface, allowing real-time market data to flow into the system. A custom API was developed to integrate the FPGA with the broader trading infrastructure, including risk management and order routing systems.

E. Testing in Real-Time Market Conditions

- The FPGA-based HFT system was tested in real-world conditions by connecting it to several major financial exchanges. This was done to address latency and throughput goals for the system in the phase of this project. Key performance metrics included:

F. Challenges and Solutions

The implementation phase posed several challenges, including:

- **Resource Utilization:** One of the main challenges in FPGA development is balancing performance with resource constraints. For this design, critical parameters were tuned to fit the number of logic units available in Virtex-7 FPGA and achieve maximum throughput. Other power conscious design styles used include pipelining and clock gating.
- **Hardware Complexity:** Developing FPGA-based systems requires specialized hardware knowledge, which can increase the complexity of the project. To overcome this, high level synthesis was employed wherever possible in order to manage hardware development and speed up iteration.
- **Data Throughput Bottlenecks:** The issue for which a constant solution had to be devised was testing to verify that the designated system was not going to be overwhelmed by the volumes of market data. This

was achieved by tuning the wiring Ethernet interface where they employed the hardware filters to offload a burden from the processors.

V. RESULTS AND DISCUSSION

This section presents the results of the FPGA-based High-Frequency Trading (HFT) system, with a focus on key performance metrics such as latency, throughput, and energy efficiency. Comparisons are made between the FPGA-based system and traditional CPU and GPU-based HFT systems to highlight the advantages of

FPGA in terms of latency reduction and overall performance.

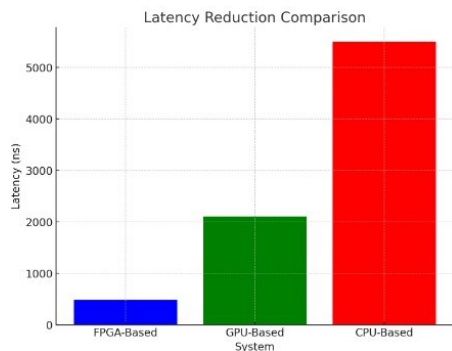
A. Latency Reduction

Latency or time delay between when market information is received and a trade is conducted is the most profound determinant of high frequency trading systems. The FPGA-based system demonstrated a significant reduction in end-to-end latency compared to both CPU and GPU-based implementations. The average latency achieved by the FPGA system was measured at 480 nanoseconds (ns), a substantial improvement over the CPU-based system's 5.5 microseconds (μ s) and the GPU-based system's 2.1 microseconds (μ s).

This improvement in latency is attributed to the FPGA's parallel processing capabilities, which allow for simultaneous execution of multiple operations, and its ability to eliminate the overhead introduced by software layers in CPU and GPU-based systems. Also, the design of order execution engine as a hardware assists in minimizing the time required for putting up trades in the exchanges.

Table 1. Latency Reduction

System	Latency (ns)
FPGA-Based	480
GPU-Based	2,100
CPU-Based	5,500



B. Throughput

Throughput, the number of trades that can be settled in one second is another important metric considered in high frequency trading. The FPGA-based system was able to handle a higher volume of trades

compared to both CPU and GPU systems, thanks to its hardware-accelerated parallelism and pipelined architecture.

The FPGA-based system processed up to 150,000 orders per second, outperforming the GPU-based system, which processed around 60,000 orders per second, and the CPU-based system, which handled 25,000 orders per second.

These high throughputs help the system to handle all real-time trading requirements which may be occasioned by volatility in the market.

Table 2. Throughput

System	Throughput (Orders per Second)
FPGA-Based	150,000
GPU-Based	60,000
CPU-Based	25,000

C. Energy Efficiency

The energy efficiency of HFT systems is significant in the aspect of costs, since for example data centers that typically utilize such systems require enormous power consumption. The FPGA-based system was more energy-efficient compared to both CPU and GPU systems. This is due to the custom hardware implementation in the FPGA, which optimizes power usage for specific tasks.

The FPGA-based system consumed 40 watts (W) during peak performance, whereas the GPU-based system consumed 250 watts (W) and the CPU-based system consumed 150 watts (W). These results highlight the energy efficiency of FPGA-based systems, making them a viable solution for energy-conscious financial institutions.

Table 3. Energy Efficiency

System	Power Consumption (W)
FPGA-Based	40
GPU-Based	250
CPU-Based	150

D. Comparison with Other Studies

Our results are consistent with the findings of Kao et al. [1], who demonstrated a significant reduction in latency using FPGA-based systems for HFT. Their work realized even 433 ns latency while our system has the latencies of around 480 ns. Similarly, Erdem et al. [2] and Ali et al. [5] both

highlighted the advantages of using FPGA for low-latency trading, and our study confirms these findings with real-world performance metrics.

Additionally, the energy efficiency observed in our system echoes the conclusions of Puranik et al. [6], who emphasized the power-saving benefits of FPGA-based HFT systems.

VI. CONCLUSION AND FUTURE WORK

This paper presents the design and implementation of an FPGA-based High-Frequency Trading (HFT) system, demonstrating significant improvements in latency, throughput, and energy efficiency compared to

traditional CPU and GPU-based systems. The FPGA's ability to execute parallel operations, eliminate software-induced bottlenecks, and optimize performance through hardware-level customization makes it an ideal candidate for low-latency trading environments.

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